optimizes for aggregation operations like sums, averages, and counts, as these calculations can be performed directly on compressed column data without decompressing entire datasets. Additionally, this approach supports efficient parallel processing, where multiple columns can be processed simultaneously across different CPU cores, making it particularly well-suited for large-scale data analytics and reporting scenarios.

The way in which timeseries data is stored as rows is shown as follows:

```
Row 1: [2024-01-15 14:30:00, temp=22.5, humidity=65, pressure=1013.2]
Row 2: [2024-01-15 14:31:00, temp=22.7, humidity=64, pressure=1013.1]
Row 3: [2024-01-15 14:32:00, temp=22.6, humidity=66, pressure=1013.3]
```

And how the same data would be stored in a columnar format:

```
Temperature:  [22.5, 22.7, 22.6, 22.8, 22.4, ...]
Humidity:     [65, 64, 66, 65, 67, ...]
Pressure:     [1013.2, 1013.1, 1013.3, 1013.0, ...]
Timestamps:   [14:30:00, 14:31:00, 14:32:00, ...]
```

4.3.3 Graph structures

Graph structures represent entities and their relationships as nodes and edges, providing natural models for interconnected systems. Digital twins often need to understand complex relationships between assets, locations, systems, and processes that graph structures handle more intuitively than relational models.

Graph structures excel at traversal queries that follow relationships across multiple entities. Finding all sensors within a building, identifying equipment dependencies, or tracing supply chain relationships becomes straightforward with graph traversal algorithms. Figure 4.7 shows a simple example of how a graph structure is used to represent a house, it's rooms, and their devices. In this example finding the smart bulbs in the house at 742 Evergreen Terrace involves finding the node that represents the address, then traversing the outbound edges with the label `hasRoom`, to get the rooms of the house, followed by traversing the outbound edges with label `hasDevice`, to find the devices, and finally looking for the nodes of type `Smart Bulb`.